

# Ketan Arora

## MECHANICAL ENGINEERING. ENERGY SYSTEMS.

### CAREER OBJECTIVES

"Having completed my Post Graduation in Mechanical Engineering, I seek to launch my career through a challenging role that tests my potential, expands my learning and provides me a valuable skillset."

### EDUCATION

**Indian Institute of Technology, Mandi.**

*M. Tech in Mechanical Engineering (Energy Systems),  
2019-21 (CGPA 8.43/10)*

**Dr. B. R. Ambedkar National Institute of Technology,  
Jalandhar.**

*B. Tech in Mechanical Engineering, 2015-19. (67.6%)*

**Gobindgarh Public School, Mandi Gobindgarh.**

(Affiliated to CBSE, New Delhi).

*Class 12 (AISSCE), 2015 (91.6%)*

*Class 10 (AISSE), 2013 (95%)*

### CONTACT

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### SKILLS

- ANSYS Fluent, Solidworks, MATLAB.
- Flexible with working environment and domains.
- Good written and oral communication skills, interpretation, analytical abilities
- Case study build up and presentation skills
- Problem Identification and Problem Solving

### TECHNICAL PROJECTS & TRAININGS

- Project on '**Experimental and numerical studies on performance of phase change material heat exchanger**'. (Aug 2020- May 2021)
- Project on '**Fabrication of superhydrophobic surfaces with the help of different types of carbon soot and adhesives and fabrication of test rig for the same**'. (Jan- May 2019) (Presented a research paper on the same)
- '**Industrial Training**' at Sonalika International Tractors Limited, Hoshiarpur (June- July 2018). (Engine Assembly Section)
- *Manufacturing Processes Training*. June 2016, NITJ.
- *Energy Systems Laboratory* (Aug 2019-Dec 2019).

### ACHIEVEMENTS & INVOLVEMENTS

- Qualified GATE 2019 (AIR 6362), GATE 2020 (AIR 4922), GATE 2021 (AIR 4027)
- Inter IIT Sports Meet 2019- IIT Mandi Football Team
- Headed NETRA (Photography Club, NIT Jalandhar)
- Qualified JEE Advanced 2015 (AIR 9198)
- State Level Badminton player at school
- Case Study (Team Event ) Winner at National Convention Student Quality Control Circles, Halol.

## COURSE WORK

- Advanced Solid Mechanics
- Advanced Thermodynamics
- Heat Transfer
- Energy Economics
- Fluid Mechanics
- Strength of Materials
- Theory of Machines
- Production and Industrial Engineering
- Material Science
- Modelling and Simulation, CAD

## TECHNICAL PROJECTS DETAILS:

- **EXPERIMENTAL AND NUMERICAL STUDIES ON PERFORMANCE OF PHASE CHANGE MATERIAL HEAT EXCHANGER'. (AUGUST 2020- MAY 2021)**
  - Tools Used: ANSYS Geometry, ANSYS Fluent, ANSYS Post-Processing, MS- EXCEL, LaTeX, MS- WORD, MS- PowerPoint, ORIGIN, Solidworks.dd
  - Self research and review of Research papers and articles.
  - Market research and ordering of relevant materials and components.
  - Fabrication and Experimentation of Lab Scale PCM based Heat Exchanger (OM-42) Setup. (2-pass and 4-pass).
  - 3-D Modelling and transient simulation of the HEXs on ANSYS modules.
  - Comparative analysis of HEX performances based on variation in orientation, number of passes, Inlet HTF Temperature, and configuration.
  - Experimental Validation of Computational Model.
  - Weekly Analysis and Report of Progress.
  - Final Thesis and Report Submission.
- **FABRICATION OF SUPERHYDROPHOBIC SURFACES WITH THE HELP OF DIFFERENT TYPES OF CARBON SOOT AND ADHESIVES AND FABRICATION OF TEST RIG FOR THE SAME'. (JAN- MAY 2019)**
  - Research on Hydrophobic materials and suitable application based base material.
  - Fabrication of Test Rig to test samples.
  - Roughness variation of base material for validation with reference results.
  - Analysis and Results.